

A literal counterexample to Tokarev's c -convex/sub- B -convex question

arXiv automatic math run fa_banach_001

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Source and Question

The source is Eugene V. Tokarev, *On sub B -convex Banach spaces*, arXiv:math/0206107, submitted 2002-06-11 [1]. Problem 1 in the source asks:

Whether any c -convex Banach space X generates a sub B -convex class X^f ?

nuclear norm $\nu_1(u)$ satisfies the estimate $\nu_1(u) \leq \lambda_X \|u\|$.

4. If X contains a subspace isomorphic to l_1 then its conjugate X^* contains a subspace isomorphic to l_2 .
5. Any operator $w : l_2 \rightarrow X$ is 2-absolutely summing; the same is true for every operator $s : l_2 \rightarrow X^*$.

The famous result of J. Pisier [1] shows the existence of the corresponding Pisier's space X for any Banach space Y of cotype 2.

From preceding sections follows that for any B -convex Banach space W of cotype 2 the space G_X - the almost ω -homogeneous space for a class $\mathfrak{Q}_W((l_1 \oplus_1 W)^f)$ is the Pisier's space for W .

Problem 1. *Whether any c -convex Banach space X generates a sub B -convex class X^f ?*

If this question has an affirmative answer then for any Banach space Z of cotype 2 the corresponding Pisier's space contains among spaces of kind G_X .

Theorem 31. *For every Banach space of cotype 2, $GL_{fin}(X)$ is finite.*

Proof. Let a Banach space Y is of cotype 2. It can be isometrically embedded in a corresponding Pisier's space X_Y , which has the same properties as was listed above. Let $T : Y \rightarrow l_2$ be an absolutely summing operator. Then T is 2-absolutely summing and, consequently, 2-integral, i.e. admits a factorizations $T = k \circ j \circ s$, where $s : X \rightarrow L_\infty(\mu)$; $k \circ j : L_\infty(\mu) \rightarrow l_2$. Because of injectivity of $L_\infty(\mu)$, s can be

Definitions Used

For a Banach space X , Tokarev writes

$$S(X) = \{p \in [0, \infty] : \ell_p <_f X\},$$

where $<_f$ denotes finite representability. The paper defines X to be c -convex if $\infty \notin S(X)$, and B -convex if $1 \notin S(X)$. The same page also notes that $S(X)$ is invariant under finite equivalence.

In Section 4, a Banach space is called sub B -convex if it is of the form

$$Y = \ell_1 \oplus_1 W$$

where W is an infinite-dimensional B -convex space. A finite-equivalence class X^f is called sub B -convex if it contains such a space Y .

Proof Intuition

The wording of Problem 1 allows X itself to be B -convex. But a sub B -convex representative always contains a literal ℓ_1 summand. Consequently its finite spectrum contains 1. This is incompatible with any B -convex finite-equivalence class. Taking $X = \ell_2$ gives the smallest possible example: it is c -convex, but its spectrum is just $\{2\}$.

Counterexample

Lemma 1. *If X^f is a sub B -convex class, then $1 \in S(X)$.*

Proof. By definition, there exists a space $Y \in X^f$ such that

$$Y = \ell_1 \oplus_1 W$$

with W infinite-dimensional and B -convex. The canonical copy $\ell_1 \oplus \{0\} \subset Y$ is isometric to ℓ_1 . Therefore $\ell_1 <_f Y$, so $1 \in S(Y)$. Since $Y \in X^f$, $X \sim_f Y$, and the spectrum is invariant under finite equivalence. Hence $1 \in S(X)$. $\square$

Theorem 1. *The Banach space $X = \ell_2$ is c -convex but X^f is not a sub B -convex class. Hence Tokarev's Problem 1 has a negative answer as stated.*

Proof. The Hilbert space ℓ_2 has type 2 and cotype 2. Using the type/cotype formula quoted in the source,

$$\inf S(\ell_2) = 2, \quad \sup S(\ell_2) = 2.$$

Thus $S(\ell_2) = \{2\}$. In particular, $\infty \notin S(\ell_2)$, so ℓ_2 is c -convex.

Suppose, toward a contradiction, that $(\ell_2)^f$ were sub B -convex. By the lemma, $1 \in S(\ell_2)$. This contradicts $S(\ell_2) = \{2\}$. Therefore $(\ell_2)^f$ is not a sub B -convex class. $\square$

Verification Notes

The proof uses only definitions and the standard finite-equivalence invariance of $S(X)$, both stated in the source paper. No computational check is involved. The key check is that a sub B -convex representative $\ell_1 \oplus_1 W$ contains ℓ_1 isometrically, forcing $1 \in S(X^f)$.

Limitations

This packet resolves only the literal wording of Problem 1. It is quite plausible that the intended question was restricted to c -convex classes which are not B -convex, equivalently to classes with

$$1 \in S(X), \quad \infty \notin S(X).$$

That intended version is not addressed here.

Novelty Check

Before promotion, the four lightweight run indexes were searched for this arXiv id, sub B -convexity, c -convexity, and Tokarev. No duplicate solution, attempt, or proof-gap packet for this target was found. Web searches on 2026-07-03 for exact phrases including

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"Whether any" "c-convex" "sub B-convex",  
"sub B-convex" "c-convex" Tokarev,  
"On sub B-convex Banach spaces" "sub B-convex class"
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found the arXiv source page but no later explicit resolution of this literal edge case.

Human Review Recommendation

Review as a likely valid counterexample to the statement exactly as printed. The main reviewer decision is interpretive: whether to count it as a genuine answer to Problem 1, or to record it as an erratum-style wording issue while keeping the non- B -convex intended variant open.

References

- [1] Eugene V. Tokarev, *On sub B -convex Banach spaces*, arXiv:math/0206107, 2002. <https://arxiv.org/abs/math/0206107>